



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

FLASH COLUMN VAPORIZER CONDENSATE PUMPS (P-2309A/B)

Docu. No. : 14780-8120-PS-P2309
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 7

TITLE :

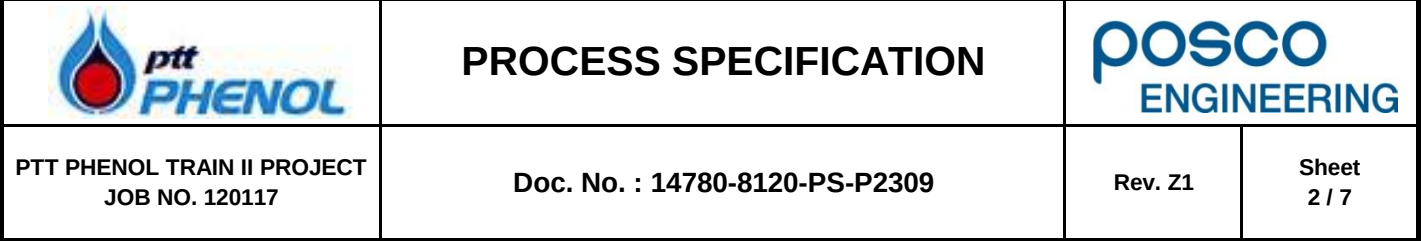
FLASH COLUMN VAPORIZER CONDENSATE PUMPS

(P-2309A/B)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-P2309
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 23, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	FEB 25, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUL 30, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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Record of Revision	
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				PROJECT SPECIFICATION																																											
UOP LLC 25 East Algonquin Road Des Plaines, Illinois 60017-5017 USA				963766 - 506 SHEET 3 / 7																																											
SEALLESS PUMPS AND DRIVERS				<table border="1" style="width:100%; border-collapse: collapse;"> <tr> <th>REV</th> <th>DATE</th> <th>BY</th> <th>APVD</th> <th>REV</th> <th>DATE</th> <th>BY</th> <th>APVD</th> </tr> <tr> <td>0</td> <td>09-Apr-12</td> <td>MW</td> <td>DRL</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>1</td> <td>31-May-12</td> <td>MW</td> <td>MEH</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>2</td> <td>14-Aug-12</td> <td>MW</td> <td>MEH</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>3</td> <td>16-Oct-13</td> <td>MW</td> <td>DRL</td> <td></td> <td></td> <td></td> <td></td> </tr> </table>				REV	DATE	BY	APVD	REV	DATE	BY	APVD	0	09-Apr-12	MW	DRL					1	31-May-12	MW	MEH					2	14-Aug-12	MW	MEH					3	16-Oct-13	MW	DRL				
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APPLICABLE TO: ☐ PROPOSALS ☐ PURCHASE ☒ AS BUILT																																															
FOR _____ UNIT _____																																															
SITE _____ SERVICE Flash Column Vaporizer Condensate Pumps																																															
NO. REQ 2 PUMP SIZE _____ TYPE Sealless Mag. Drive NO. STAGES 1																																															
MANUFACTURER _____ MODEL _____ SERIAL NO. _____																																															
DIMENSIONS PER: ☐ SI UNITS ☐ US UNITS EQUIPMENT TO COMPLY WITH APPLICABLE: ☐ ISO ☐ US STANDARDS																																															
NOTES: INFORMATION BELOW TO BE COMPLETED: ☐ BY PURCHASER ☐ BY MANUFACTURER ☒ BY MANUFACTURER OR PURCHASER																																															
☐ GENERAL (5.15.1.1)																																															
PUMPS TO OPERATE IN (PARALLEL) _____ NO. MOTOR DRIVEN 2 NO. TURBINE DRIVEN _____																																															
(SERIES) WITH _____ PUMP ITEM NO. P-2309A/B PUMP ITEM NO. _____																																															
GEAR ITEM NO. _____ MOTOR ITEM NO. _____ TURBINE ITEM NO. _____																																															
GEAR PROVIDED BY _____ MOTOR PROVIDED BY _____ TURBINE PROVIDED BY _____																																															
GEAR MOUNTED BY _____ MOTOR MOUNTED BY _____ TURBINE MOUNTED BY _____																																															
GEAR DATA SHT. NO. _____ MOTOR DATA SHT. NO. _____ TURBINE DATA SHT. NO. _____																																															
☐ OPERATING CONDITIONS (6.1.3)																																															
CAPACITY, NORMAL 12.4 m ³ /h RATED 13.3 m ³ /h																																															
OTHER _____ 5.45 m ³ /h																																															
SUCTION PRESS MAX./RATED 7.256 / 2.485 kg/cm ² [g]																																															
DISCHARGE PRESSURE 4.066 kg/cm ² [g]																																															
DIFFERENTIAL PRESSURE 1.58 kg/cm ²																																															
DIFF. HEAD 17.0 m NPSHA 2.5 m																																															
PROCESS VARIATIONS (7.1.2) _____																																															
STARTING CONDITIONS (7.1.3) _____																																															
SERVICE: ☒ CONT. ☐ INTERMITTENT (STARTS/DAY) _____																																															
☐ PARALLEL OPERATION REQ'D (6.1.12)																																															
☐ SITE DATA																																															
LOCATION: (6.1.29)																																															
☐ INDOOR ☐ HEATED ☐ UNDER ROOF																																															
☒ OUTDOOR ☒ UNHEATED ☐ PARTIAL SIDES																																															
☒ GRADE ☐ MEZZANINE ☐ _____																																															
☐ ELECTRIC AREA CLASSIFICATION (6.1.24)																																															
CL 1 GR IIA DIV ZONE 2																																															
☐ WINTERIZATION REQ'D ☐ TROPICALIZATION REQ'D. (7.2.4.5)																																															
SITE DATA (6.1.29) SEE GENERAL NOTES ON SHEET 6 *																																															
☐ ALTITUDE 0 BAROMETER _____																																															
☐ RANGE OF AMBIENT TEMPS: MIN/MAX. _____ / _____ 0																																															
☐ RELATIVE HUMIDITY: MIN / MAX _____ / _____ %																																															
UNUSUAL CONDITIONS: (6.1.29) ☐ DUST ☐ FUMES																																															
☐ OTHER _____																																															
☐ LIQUID																																															
TYPE OR NAME OF LIQUID Condensate																																															
PUMPING TEMP °C NORMAL 135 MAX. _____ MIN. _____																																															
VAPOR PRESSURE 3.28 kg/cm ² [a] @ 135 °C																																															
TEMPERATURE-VAPOR PRESSURE CURVE NO: _____																																															
RELATIVE DENSITY (SPECIFIC GRAVITY):																																															
NORMAL 0.931 MAX _____ MIN _____																																															
SPECIFIC HEAT, Cp _____ 1.02 kcal/(kg.°C)																																															
VISCOSITY 0.204 cP @ 135 °C																																															
MAX. VISCOSITY _____ (cP) @ _____ °C																																															
TEMPERATURE-VISCOSITY CURVE NO: _____																																															
THERMAL EXPANSION: _____ mm/m/°C																																															
THERMAL CONDUCTIVITY: _____ 0.587 kcal/(h.m.°C)																																															
CORROSIVE/EROSIVE AGENT _____ (6.11.7)																																															
CORROSION ALLOWANCE (6.3.3) _____ mm																																															
☐ CHLORIDE PRESENT: CONCENTRATION (PPM) _____ (7.3.2.6)																																															
☐ WET H2S PRESENT: CONCENTRATION (PPM) _____ (6.11.11)																																															
☐ SOLIDS PRESENT (6.2.1): CONCENTRATION (% by wt.) _____																																															
PARTICLE SIZE _____ HARDNESS _____																																															
SIZE DISTRIBUTION _____																																															
LIQUID (6.1.3) ☐ HAZARDOUS ☐ FLAMMABLE																																															
POLYMERIZATION CHARACTERISTICS (6.2.1): _____																																															
☐ OTHER _____																																															
REMARKS: _____																																															
☐ UTILITY CONDITIONS: SEE GENERAL NOTES ON SHEET 6 *																																															
STEAM: DRIVERS HEATING																																															
MIN _____ _____ _____ 0																																															
MAX _____ _____ _____ 0																																															
ELECTRICITY: DRIVERS HEATING CONTROL SHUTDOWN																																															
VOLTAGE _____ _____ _____																																															
HERTZ _____ _____ _____																																															
PHASE _____ _____ _____																																															
COOLING WATER: (6.1.22) SOURCE																																															
TEMP. INLET _____ 0 MAX. RETURN _____ 0																																															
PRESS. NORM _____ 0 DESIGN _____ 0																																															
MIN RETURN _____ 0 MAX ALLOW DP _____																																															
FOULING FACTOR _____ 0																																															
CHLORIDE CONCENTRATION (PPM) _____ (7.3.2.6)																																															
INSTRUMENT AIR: MAX/MIN PRESS _____ / _____ 0																																															
☐ SYSTEM DESCRIPTION (6.2.2)																																															
SUCTION VESSEL: ☐ VENTED TO ATMOS ☐ CLOSED																																															
PUMP LOCATION: ☐ BELOW LIQ. LEVEL ☐ ABOVE LIQ. LEVEL																																															
SUCTION VESSEL ON LEVEL CONTROL? ☐ YES ☐ NO																																															
PRESSURE SENSOR ON SUCTION VESSEL ☐ YES ☐ NO																																															
SUCTION VESSEL PRESSURE MAINTAINED BY: _____																																															
IF FLUID LEVEL OR TANK PRESSURE DROPS TOO LOW, WILL																																															
SYSTEM AUTOMATICALLY STOP THE PUMP? ☐ YES ☐ NO																																															
WILL THE PUMP RUN DRY IN NORMAL OPERATION? ☐ YES ☐ NO																																															
REMARKS: See Pump Note 1, 3, 5 and C2																																															
See Motor Note M10.																																															
ESTIMATED SHUTOFF PRESSURE 9.233 kg/cm ² [g]																																															



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SEALLESS PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 506

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	31-May-12	MW	MEH				
2	19-Jul-12	MW	DRL				
3	25-Sep-13	MW	CJH				

REV	PERFORMANCE	CONSTRUCTION - CONT'D																		
	PROPOSAL CURVE NO. ☐ RPM 2950 ☐ IMPELLER DIA.: RATED _____ MAX _____ MIN _____ mm ☐ HYD. EFFICIENCY AT RATED _____ 51 _____ % ☐ HYSTERESIS & MECH. LOSSES _____ 0.7 kW _____ kW ☐ OVERALL EFF. % 37.8 RATED POWER _____ 1.5 _____ kW ☐ MAX POWER @ RATED IMPELLER _____ kW ☐ MINIMUM CONTINUOUS FLOW: THERMAL _____ m³/h STABLE _____ m³/h ☐ ROTOR CHAMBER TEMP. RISE OPERATING (6.2.1) _____ °C ☐ ROTOR CHAMBER TEMP. RISE ON SHUTDOWN _____ °C ☐ PREFERRED OPER. REGION _____ TO _____ m³/h ☐ ALLOWABLE OPER. REGION _____ TO _____ m³/h LIMITED BY: ☐ VIBRATION ☐ OTHER _____ ☐ MAX HEAD @ RATED IMPELLER _____ m ☐ PERCENT RISE TO SHUTOFF (6.1.12) _____ ☐ ORIFICE USED TO STEEPEN CURVE OR GIVE CONT. RISE (6.1.12) ☐ NPSHR AT RATED CAPACITY (6.1.19) _____ m ☐ MAX SUCTION SPECIFIC SPEED ALLOW (6.1.10) 13,000 max (m³/hr, meters) ☐ SUCTION SPECIFIC SPEED QUOTED _____ ☐ MAX SOUND PRESS LEVEL REQ'D (6.1.15) _____ dBA ☐ EST. MAX SOUND PRESS LEVEL _____ dBA REMARKS _____	NOZZLE CONDITIONS: (6.4.2) <table border="1"> <thead> <tr> <th>SIZE (NPS)</th> <th>FLANGE CLASS</th> <th>FAC'G</th> <th>POSITION</th> </tr> </thead> <tbody> <tr> <td></td> <td>150</td> <td>RF</td> <td></td> </tr> <tr> <td></td> <td>150</td> <td>RF</td> <td></td> </tr> </tbody> </table> SUCTION _____ DISCHARGE _____ PRESSURE CASING AUX. CONNECTIONS: (6.4.3) <table border="1"> <thead> <tr> <th>NO.</th> <th>SIZE (NPS)</th> <th>TYPE</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> <td>Provide with Valve</td> </tr> </tbody> </table> ☐ DRAIN ☐ VENT ☐ PRESSURE GAUGE ☐ TEMP GAUGE ☐ WARM-UP ☐ EXT. FLUSH/LEAK-OFF ☐ PURGE/FLUSHOUT VENDOR SUPPLY VALVES FOR: (7.3.2.3) ☐ VENTS(S) ☐ DRAIN(S) ☐ ROTOR CAVITY DRAINABLE THRU SECONDARY DRAIN (6.1.8) ☐ CYLINDRICAL THREADS REQ'D FOR AUX. CONN'S (6.4.3.8.3) ☐ FLANGE THICKNESS REQ'S NON-STD BOLT LENGTH (6.4.2.3) ☐ THD CONN. TO SECONDARY CASING IS NOT ACCEPT. (6.4.3.8) SECONDARY CASING HAS: (6.8.8) ☐ DRAIN CONN. ☐ & FLUSH PROVISION	SIZE (NPS)	FLANGE CLASS	FAC'G	POSITION		150	RF			150	RF		NO.	SIZE (NPS)	TYPE			Provide with Valve
SIZE (NPS)	FLANGE CLASS	FAC'G	POSITION																	
	150	RF																		
	150	RF																		
NO.	SIZE (NPS)	TYPE																		
		Provide with Valve																		
	CONSTRUCTION APPLICABLE STANDARD: ☐ API 685 ☒ OTHER ASME B73.3 (SEE REMARKS) ROTATION: (VIEWED FROM COUPLING END) ☐ CW ☐ CCW PUMP TYPE: ☒ OH1 ☐ OH2 ☐ OH5 ☐ OTHER _____ CASING MOUNTING: ☐ CENTERLINE ☐ NEAR CENTERLINE ☒ FOOT ☐ IN-LINE ☐ OTHER _____ CASING TYPE: ☒ SINGLE VOLUTE ☐ MULTIPLE VOLUTE ☐ DIFFUSER CASE PRESSURE RATING: ☐ MAX ALLOWABLE WORKING PRESSURE 14.06 kg/cm² g @ _____ 0 ☐ HYDROTEST PRESSURE 21.09 kg/cm² g DESIGN PUMP FOR: ☐ FULL ☐ _____ kg/cm² g VACUUM REMARKS _____	AUXILIARY PIPING (7.3) ☐ AUXILIARY PIPING PLAN: 23-S CIRC. PIPING MATERIALS: ☐ TUBING ☐ CARBON STEEL ☐ PIPE ☐ STAINLESS STEEL CIRC. PIPING ASSEMBLY: (7.5.2.10) ☐ THREADED ☐ UNIONS ☐ SOCKET WELDED ☐ FLANGED ☐ TUBE FITTINGS ☐ COOLING WATER PIPING PLAN (7.3.4) M-S C.W. PIPING ASSEMBLY: ☐ PIPE ☐ TUBING FITTINGS _____ C.W. PIPING MATERIALS: ☐ STAINLESS STEEL ☐ C. STEEL ☐ GALVANIZED COOLING WATER REQUIREMENTS: ☐ JACKET ☐ BEARING HOUSING _____ m³/h HEAT EXCHANGER _____ m³/h TOTAL COOLING WATER _____ m³/h STEAM PIPING: ☐ TUBING ☐ PIPE ☐ OTHER REMARKS _____																		
	ROTOR: ☐ OPEN ☐ SEMI-OPEN ☐ FAB'D IMPELLERS ARE ACCEPT ☐ RENEWABLE CASE WEAR RINGS ARE NOT REQUIRED (6.7.1) ☐ RENEWABLE IMP. WEAR RINGS ARE NOT REQUIRED (6.7.1) ☐ NON-GROOVED BEARINGS ARE ACCEPTABLE (6.10.5) ☐ SHAFT DIAMETER BETWEEN BEARINGS _____ mm ☐ SPAN BETWEEN BEARING CENTERS _____ mm ☐ SPAN BETWEEN BEARING & IMPELLER _____ mm REMARKS _____	MATERIALS ☐ APPENDIX H CLASS (6.11.1) _____ ☐ MIN DESIGN METAL TEMP (6.11.4.5) 15 °C ☐ CASE Killed Carbon Steel IMPELLER Killed Carbon Steel ☐ CASE/IMPELLER WEAR RINGS _____ ☐ SHAFT 316 Stainless ☐ CONTAINMENT SHELL/STATOR LINER 316 Stainless ☐ INNER MAG. SHEATHING/ROTOR LINER _____ ☐ BEARINGS SiC ☐ OTHER _____																		



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SEALLESS PUMPS AND DRIVERS

PROJECT SPECIFICATION

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				

REV	SECONDARY CONTROL/CONTAINMENT	MAG-DRIVE SPECIFIC - CONT'D																			
	☐ SECONDARY CONTROL (6.8.1) ☐ MAX LEAKAGE ON PRIMARY FAILURE: _____ m ³ /h ☐ SECONDARY CONTAINMENT (6.8.1) AUX. SEAL DEVICE: ☐ MECHANICAL SEAL ☐ BUSHING ☐ OTHER _____ DEVICE IS: ☐ DRY RUNNING ☐ WET RUNNING ☐ INSIDE ☐ OUTSIDE OF BEARINGS (MAG-DRIVE) ☐ OTHER: _____ MECHANICAL SEAL DATA: ☐ APPENDIX H SEAL CODE _____ ☐ SEAL MANUFACTURER _____ ☐ SIZE AND TYPE _____ / _____ ☐ MANUFACTURER CODE _____ ☐ REMARKS _____	SHAFT COUPLINGS: (DRIVER-TO-PUMP) ☐ MAKE _____ ☐ MODEL Flexible with SS Disks ☐ NON-LUBE ☐ LUBRICATED: (SEE REMARKS) ☐ RATING (kW/100 RPM) _____ ☐ SERVICE FACTOR _____ ☐ LIMITED END FLOAT REQUIRED _____ ☐ SPACER, LENGTH _____ mm ☐ NOT REQUIRED ☐ MATERIAL: HUBS _____ SPACER _____ ELEMENTS _____ ☐ SHAFT DIAMETER AT COUPLING _____ mm DRIVER HALF COUPLING MOUNTED BY: ☐ PUMP MFR. ☐ DRIVER MFR. ☐ PURCHASER REMARKS _____ SEPARATE MOTOR DRIVE APPLICABLE SPEC: ☐ IEEE841 ☐ API541 ☐ OTHER: _____ ☐ FRAME _____ ☐ ENCLOSURE _____ ☐ LUBE _____ BEARING (TYPE/NUMBER): ☐ RADIAL _____ / _____ ☐ THRUST _____ / _____ REMARKS _____	1																		
	MAG-DRIVE SPECIFIC ☐ DESIGN FOR REMOVAL OF DRIVE END WITHOUT DISTURBING THE PRESSURE CASING OR DRIVER (9.1.1.4) MAGNETIC COUPLING: ☐ SYNCHRONOUS ☐ TORQUE RING MAGNETS: <table border="1"> <thead> <tr> <th></th> <th>OUTER</th> <th>INNER</th> </tr> </thead> <tbody> <tr> <td>☐ MAG. MATERIAL</td> <td></td> <td></td> </tr> <tr> <td>☐ MOUNT. METHOD</td> <td></td> <td></td> </tr> <tr> <td>☐ TEMP. LIMIT. °C</td> <td></td> <td></td> </tr> <tr> <td>☐ HERMETIC SEALED</td> <td></td> <td></td> </tr> <tr> <td>☐ NO. OF MAGNETS</td> <td></td> <td></td> </tr> </tbody> </table> ☐ SOFT START REQ'D ☐ HARD START CAPABILITY ☐ TORQUE RATING (DECOUPLING), Nm _____ ☐ MAX TORQUE REQ'D ON STARTING, Nm (9.1.3.7a) _____ ☐ PUMP TORQUE AT RATED (+5%), Nm (9.1.3.7b) _____ ☐ ACTUAL SERVICE FACTOR (9.1.3.8) _____ % ☐ TORQUE REQ'D FOR FULL CURVE, Nm (9.1.3.7c) _____ ☐ DESIGN FOR FULL CURVE TORQUE REQUIREMENTS (9.1.3.7c) ☐ SUBMIT MAG-COUPLING TORQUE - TEMP CURVE (9.1.3.9) ☐ SUBMIT SPEED VS TORQUE CURVE (9.1.3.10) ☐ _____ EXTERNAL BEARING (TYPE/NUMBER): ☐ RADIAL _____ Ball _____ / _____ ☐ THRUST _____ Ball _____ / _____ LUBRICATION: (9.1.4.3) ☐ RING OIL ☒ FLINGER ☒ FLOOD ☐ PURGE OIL MIST ☒ PURE OIL MIST ☒ OTHER _____ ☐ CONSTANT LEVEL OILER PREFERENCE (SEE REMARKS) (9.1.4.2.1) ☒ OIL VISC. ISO GRADE _____		OUTER	INNER	☐ MAG. MATERIAL			☐ MOUNT. METHOD			☐ TEMP. LIMIT. °C			☐ HERMETIC SEALED			☐ NO. OF MAGNETS			BASEPLATES (9.1.5.3): ☐ API BASEPLATE NUMBER _____ (APPENDIX M) ☐ NON-GROUT CONSTRUCTION: (9.1.5.3.13) ☐ BASEPLATE MATERIAL _____ ☐ BOLT V-INLINE PUMP TO PAD/FOUNDATION (6.1.30.3) REMARKS _____	2
	OUTER	INNER																			
☐ MAG. MATERIAL																					
☐ MOUNT. METHOD																					
☐ TEMP. LIMIT. °C																					
☐ HERMETIC SEALED																					
☐ NO. OF MAGNETS																					
		CANNED MOTOR SPECIFIC ☐ SOLID MEDIA POTTING ALLOWED IN STATOR (9.2.2.2) ☐ CERAMIC INSULATION ACCEPTABLE (9.2.2.8) ☐ DESIGN MOTOR FOR FREQUENT STARTS (9.2.2.9) ☐ _____ STARTS PER _____ ☐ IMPACT ON LIFE: (REMARKS) ☐ DESIGN MOTOR FOR: (9.2.2.9) ☐ TWO-SPEED ☐ VARIABLE SPEED _____ ☐ UL, FM OR EQUIVALENT REQ'D (9.2.2.10) ☐ UL ☐ FM ☐ OTHER ☐ CERTIFICATION OF IEEE 252 TEST REQUIRED (9.2.7) ☐ DECONTAMINATION CONNECTION ON STATOR (9.2.2.11) REMARKS _____	3																		
	ADDITIONAL REMARKS: Pumps shall comply with UOP Standard Specifications 7-12 and 5-16		4																		
			5																		
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1	08-May-12	MW	DRL				

REV

GENERAL NOTES:

- Vendor's offering shall be a unit of duplicate size and design which has a successful record of proven field service at operating conditions similar to those specified (proto-types are not acceptable). An installation list shall be submitted upon request. In the event no similar unit is available, the vendor may offer an alternate with a detailed explanation of where the offering differs from his field proven equipment.
- The design pressure of downstream equipment, piping, and instrumentation has been arrived at by adding 1.25 times the rated pump differential to the maximum suction pressure. Contractor to adjust as necessary or verify that downstream equipment, piping, and instrumentation design pressures are adequate for the shutoff pressure of purchased pumps.
- Flange classes are specified in accordance with ASME B16.1 and ASME B16.5. Flange classes listed in the UOP Project Specifications are based upon design pressure and temperature conditions only, and do not account for other loads. The final design of all flanges shall account for gasket seating and external loads. Differential thermal expansion for dissimilar joints and transient thermal conditions such as start-up/shutdown and operational upset shall be accommodated.
- All flanges intended for use with spiral wound gaskets shall have a flange surface finish of 125 Ra minimum to 250 Ra maximum. Finishes shall be judged by visual comparison with surface finish roughness standards conforming to ASME B46.1.
- Testing
 - a) Each pump shall have a factory performance test.
 - b) Pressure containing parts (including auxiliaries) shall be tested hydrostatically with liquid at a minimum of 1-½ times the maximum allowable working pressure.
- A self-cleaning type strainer shall be built into the pump discharge nozzle to prevent grit and other large size particles from passing through the flushing line and into the motor section.
- Alarms and Shutdowns
The following alarm and shutdown transmitters shall be provided:

	Alarms	Shutdown
Low discharge flow (by others)		
High/low motor power	X	X
High/low stator pressure (Canned type only)	X	
- Shutdown Circuitry shall be designed to stop the operating pump for an abnormal condition by any of the listed shutdown variables.
- Start circuitry shall be designed with a time delay to maintain control circuit energization until the discharge flow is established.
- Alarm circuitry shall be designed to series connect all listed alarm contacts, for each pump service, to activate a common trouble alarm (by others) in the control center.
- Contractor shall coordinate the details of the pump shutdown/start/alarm circuitry.

G1. Lubrication system is applied to high voltage motor's and all pumps' bearings except for vertical and diaphragm pumps.

* SEE BASIC ENGINEERING DESIGN QUESTIONNAIRE (BEDQ), ENGINEERING DESIGN INFORMATION (EDI) PROJECT 963763 A.4, SECTIONS 3 AND 4.

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SEALLESS PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 506

SHEET 7 / 7

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	16-Oct-13	MW	DRL				

REV

The following notes apply to those services for which they are specified in this Project Specification:

PUMP NOTES:

1. A spill back to suction source will be provided to insure satisfactory pump operation at minimum continuous flow. Pump vendor to advise minimum continuous capacity.
3. Normal, rated and other capacities include 2.27 m³/hr through restriction orifice-in spillback line.
5. The "capacity, other" is based on 50% of the lowest normal operating flow.

- C1. Variation in pumped fluid physical properties is listed below (Item No. P-2308A/B)

Temperature, °C	116.7	138.9	163.9
Vapor pressure, kg/cm ² [a]	1.88	3.573	6.954
Viscosity, cP	0.239	0.198	0.166
Specific Heat, kcal/(kg.°C)	1.012	1.024	1.041
Thermal Conductivity, kcal/(h.m.°C)	0.586	0.587	0.583

- C2. Variation in pumped fluid physical properties is listed below (Item No. P-2309A/B)

Temperature, °C	135.5	160.5	185.5
Vapor pressure, kg/cm ² [a]	3.28	6.394	11.601
Viscosity, cP	0.204	0.17	0.145
Specific Heat, kcal/(kg.°C)	1.021	1.038	1.061
Thermal Conductivity, kcal/(h.m.°C)	0.587	0.584	0.387

MOTOR NOTES:

- M10. Motor and coupling shall have the torque capability to accelerate the pump to operating speed with the discharge valve open, and the discharge check valve closed against the following static pressure:

Item Number	Minimum Autostart Pressure
P-2308A/B	2.25 kg/cm ² (g)
P-2309A/B	3.80 kg/cm ² (g)

- M12. Deleted.